

Design and Implementation of a **Rule-Based** Artificial Intelligence System for the Optimization of **Personalized** Study Routines Among **Software Engineering** Students

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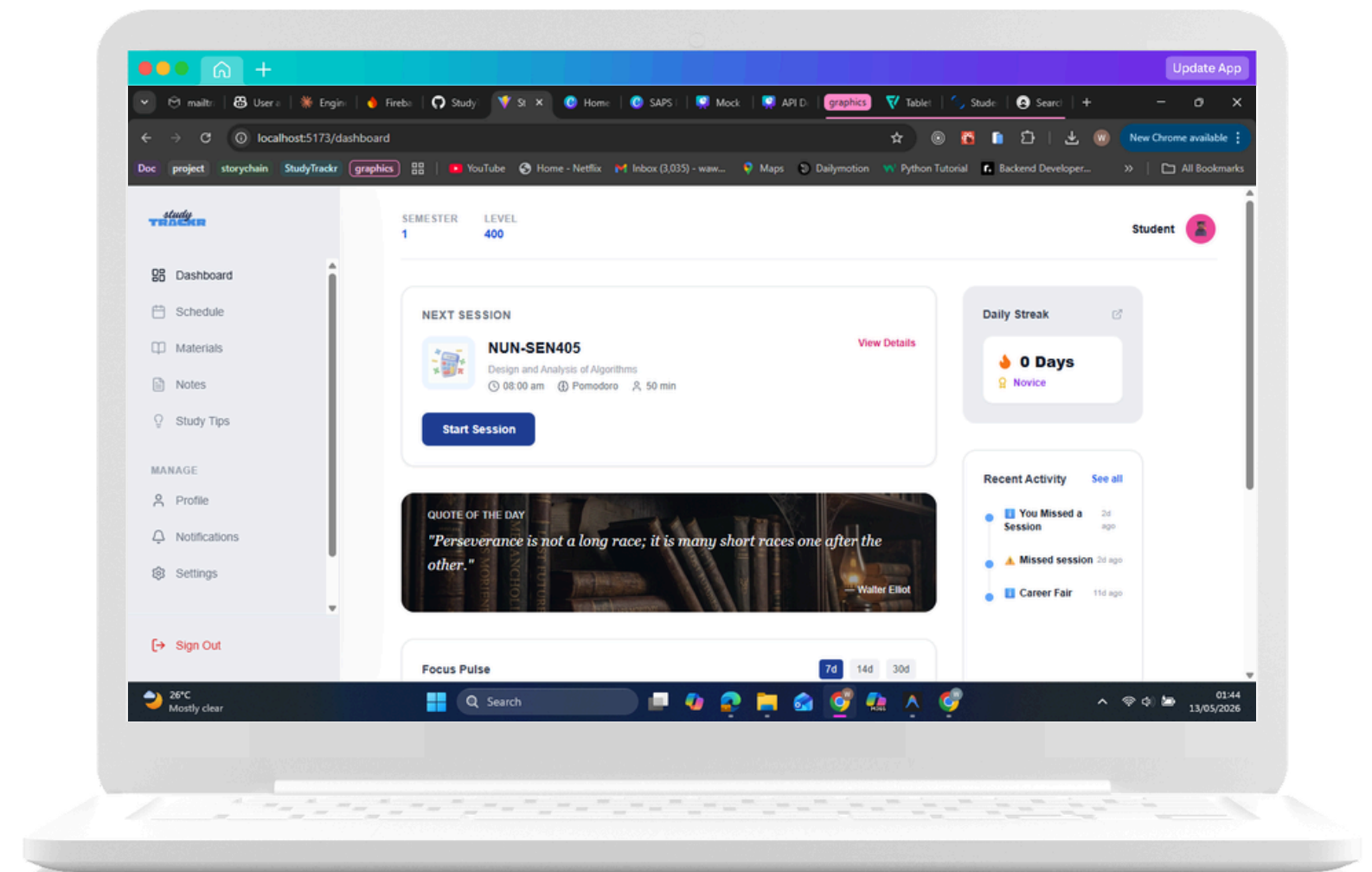
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Introduction

StudyTrackr is a web-based study routine optimisation platform designed for Nile University Software Engineering students. It uses a rule-based inference engine to generate personalised weekly study schedules tailored to each student's cognitive profile, learning preferences, and academic workload. The system adapts over time by analysing logged session data, refining its recommendations based on the student's study patterns and performance feedback, replacing unstructured improvisation with a structured, science-backed routine.



INTRODUCTION

Problem Statement

Software Engineering students manage multiple courses, deadlines, and responsibilities simultaneously, which make structured study routines difficult to maintain. A preliminary survey was conducted among Nile University Software Engineering students. It yielded a total of **61** responses and the results revealed that:

- 01 **72.1%** rely on mental notes rather than any structured tracking tool.
- 02 **41%** identified procrastination as their biggest academic hindrance.
- 03 Only **18%** reported being consistent with their study habits.

This confirms a clear execution gap, showing that a lot of students have the intent to study but lack the structure to follow through. It also outlines two root causes namely **limited awareness** of effective study techniques and **metacognitive failure**.



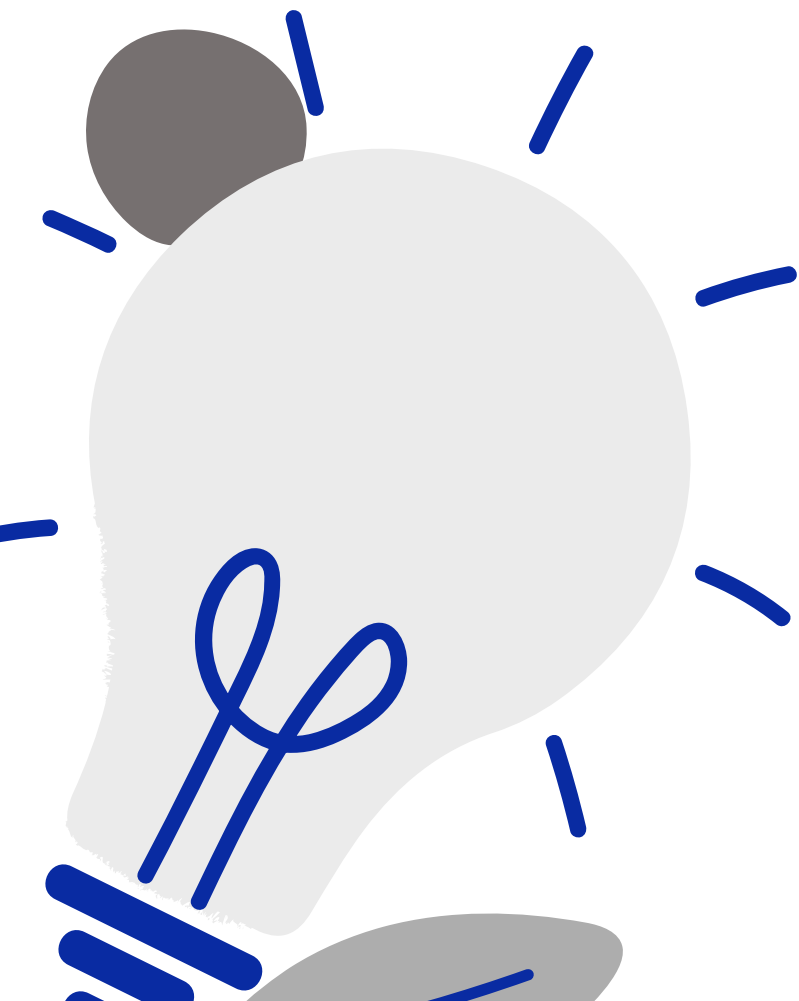
AIM AND OBJECTIVES

Aim

The aim of this project is to design and implement a functional, adaptive study routine optimization platform for Software Engineering students, which will generate personalized schedules and dynamically refine them based on study habits of the student through a rule-based inference engine.

Objectives

- ✓ To design a functional, adaptive study routine optimization platform for Software Engineering students that utilizes a rule-based inference engine and personalized learning profiles.
- ✓ To develop the designed study routine optimization platform effectively and establish a database to manage academic curricula and dynamic scheduling data on the system.
- ✓ To test the implemented study routine optimization platform to ensure that the system accurately generates comprehensive schedules and effectively refines them based on student performance feedback.



	TITLE OF PAPER/YEAR	AUTHORS	METHODOLOGY	SUMMARY OF RESULT	LIMITATION
1	Analysis of Emerging Trends in AI for Education in Nigeria (2024)	B. Bali et al.	Comprehensive review of digital libraries.	Logic-based AI is more sustainable in Nigeria due to bandwidth constraints.	Does not offer a specific implementation framework or technical blueprint.
2	The Science of Effective Learning with Spacing and Retrieval Practice (2022)	S. K. Carpenter et al.	Systematic review of educational settings.	Spacing and retrieval are high-utility but chronically underused by students.	It is strictly psychological, providing no guidance on building a tool to implement these theories.
3	A Meta-Analysis of Ten Learning Techniques (2021)	G. M. Donoghue & J. Hattie	Meta-analytic synthesis of hundreds of studies.	Distributed practice and practice testing are the most robust learning techniques.	High-level synthesis which lacks guidance on automating these techniques within a scheduling algorithm.
4	Predicting Student Performance Using Rule-Based Systems (2020)	P. Kumar et al.	Rule-based expert system methodology using if-then logic structures.	Deterministic rule-based logic is effective and transparent in educational environments.	Focuses on predicting final grades rather than real-time daily schedule optimisation.

	TITLE OF PAPER/YEAR	AUTHORS	METHODOLOGY	SUMMARY OF RESULT	LIMITATION
5	Rule-Based Expert Systems for Supporting University Students (2014)	G. Engin et al.	Design and testing of expert systems using Oracle Policy Automation.	Rule-based systems reduce administrative workload and provide students with auditable academic suggestions.	Specific to one university and not integrated with productivity tools like focus timers.
6	Anti-Procrastination Online Tool for Graduate Students Based on the Pomodoro Technique (2020)	K. Almalki et al.	Four-week diary study with 15 graduate students.	Pomodoro-based tool reduced procrastination in over 70% of participants.	Very small sample size; focused exclusively on graduate students.
7	AI-Powered Smart Study Planner: Enhancing Personalized Learning (2025)	G. Lingshetti et al.	AI-driven framework incorporating user preferences and course difficulty.	Intelligent scheduling leads to a measurable increase in student productivity.	Lacks a robust feedback loop and gamification to keep students disciplined and engaged.
8	Personalized Gamification: A Technological Approach for Student Education	S. D. Ristianto et al.	Systematic literature review of gamification trends.	Personalising gamification leads to significantly higher student retention and motivation.	It is a high-level review that provides no specific backend implementation for an XP calculation service.

SIGNIFICANCE

Student Impact

- 01 Replaces unstructured improvisation with a data-driven, personalised study framework.
- 02 Eliminates metacognitive failure by automating planning decisions based on each student's cognitive profile.
- 03 Reduces daily decision fatigue, thus freeing mental energy for actual studying rather than planning.



SIGNIFICANCE

Institutional and Academic Value

- 01 Provides the department with insights into student study consistency, performance trends, and learning preferences across levels.
- 02 Enables proactive intervention by identifying difficult courses before examination scores reveal the damage.
- 03 Serves as a reference for future research in intelligent learning systems, adaptive scheduling, and academic productivity tools.





Thank You